

Planning and Reinforcement Learning in AI Systems

(PARLAI)

Multi-agent AI (part 1)

Sarah Keren

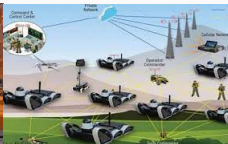
The Taub Faculty of Computer Science
Technion - Israel Institute of Technology

Multi-Agent AI

Challenges in multi-agent coordination

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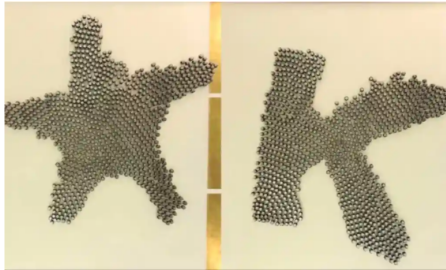
Challenges in multi-agent coordination

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Harvard's 1,000 Kilobot swarm demonstrates the future of robotics

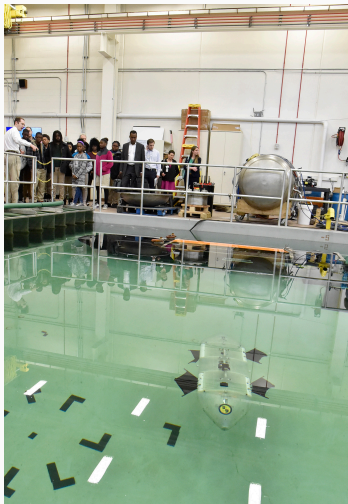
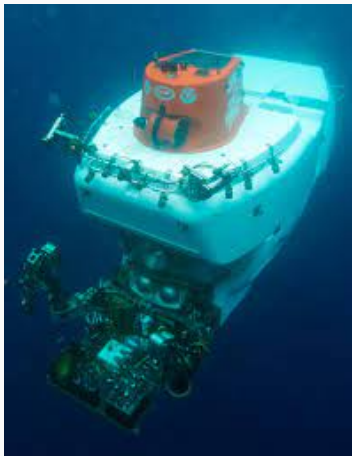
A new system has replicated the hive mind with over 1,000 simple robots collaborating without human control



Harvard's new Kilobots swarm and cooperate in a 1,000-strong mass to create complex shapes with no micromanagement. Photograph: Harvard Photograph: Harvard

From Radhika Nagpal's lab.

Challenges in multi-agent coordination



Naval Research Lab

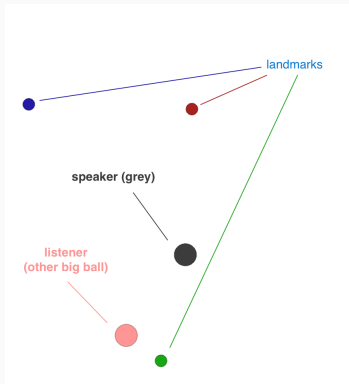
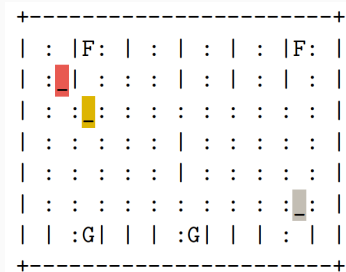
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Challenges in multi-agent coordination

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Challenges in multi-agent coordination

- Computational complexity and generality (even in full communication and centralized control settings)
- Communication Limitations
 - Low bandwidth
 - Unpredictably lossy networks
- Joint planning
 - Efficient coordination
 - Imperfect information



Why Multiple Agents?

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Why Multiple Agents?

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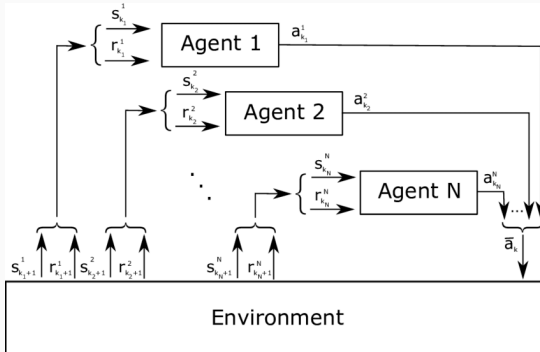
- **Parallelism:** Many robots can accomplish the task faster
- **Redundancy:** Hazardous environment with chances of losing robots
- **Required:** Too difficult to do with a single size robot
- **Complex Tasks:** Need several specialized robots
- **Real-time Requirements:** Monitor large areas, respond quickly

Sometime it's simply a constraint imposed by the environment.

The Multi-agent-Environment Interface

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To model a multi-agent setting, We will use a *Markov game*, or *stochastic game* which is a generalization of the MDP to multi-agent settings (Littman'94).

Image by Michele Chincoli and Antonio Liotta

Multi-Agent Markov Decision Process - Markov Games

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A **Markov game** is defined by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ where

- $\mathcal{S}$ is a finite set of states
- $\mathcal{A} = \{\mathcal{A}^i\}_{i=1}^n$ are the **joint actions**: a collection of action sets $\mathcal{A}^i$, one for each agent in the environment. At each timestep t , each agent i chooses an action $a_t^i \in \mathcal{A}^i$. The actions of all N agents are combined to form a **joint action** $\mathbf{a}_t = [a_t^0, \dots, a_t^N]$.
- $\mathcal{P}$ describes the probability distribution over next states when a joint action is performed and produces a transition in the environment, where $\mathcal{P}_{s,s'}^{\mathbf{a}} = \mathcal{P}[S_{t+1} = s' | S_t = s, A_t = \mathbf{a}_t]$ describes the probability of ending at state s' when the joint action $\mathbf{a}_t \in \mathcal{A}$ is performed at state s .
- $\mathcal{R} = \{\mathcal{R}^i\}_{i=1}^n$ is a collection of rewards functions $\mathcal{R}^i$ defining the reward $r^i(a_t, s_t)$ each agent receives when the joint action a_t is performed at state s_t ,
- γ is a discount factor $\gamma \in [0, 1]$.

Multi-Agent Markov Decision Process - Markov Games

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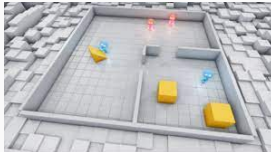
What about partially observability?

Multi-Agent Markov Decision Process - Markov Games

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In partially observable environments the i th agent can only view a portion of the true state, s_t^i .



Reminder: Partially Observable Markov Decision Process (POMDP)

A **Partially Observable Markov Decision Process**(POMDP) is a tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma, \Omega, \mathcal{O}, \beta_0 \rangle$ where

- $\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}$ and γ are as for an MDP.
- Ω is a set of observations (observation tokens),
- $\mathcal{O}$ is a sensor function specifying the conditional observation probabilities $\mathcal{O}_{s,a}^o = \mathcal{P}[O_{t+1} = o | S_t = s, A_t = a]$ of receiving observation token $o \in \Omega$ in state s after applying action a ¹.
- β_0 the initial belief: a probability distribution over the states such that $\beta_0(s)$ stands for the probability of s being the true initial state.

¹alternatively: $\mathcal{O}_s^o = \mathcal{P}[O_t = o | S_t = s]$

- A **Multi-Agent POMDP** can be defined by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma, \Omega, O, \mathfrak{B}_0 \rangle$ where
 - $\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}$ and γ are as in the definition of the Markov Game,
 - $\Omega = \{\Omega^i\}_{i=1}^n$ are the **joint observations**: a collection of observation token sets Ω^i , one for each agent in the environment. At each timestep t , each agent i observes an observation $o_t^i \in \Omega^i$. The observations of all N agents are combined to form a **joint observation** $\mathbf{o}_t = [o_t^0, \dots, o_t^N]$.
 - O is the **sensor function** describing the probability distribution over next joint observations when a joint action is performed in some state, where
$$O_{s,\mathbf{a}}^\mathbf{o} = \mathcal{P}[\mathbf{O}_{t+1} = \mathbf{o} | S_t = s, A_t = \mathbf{a}_t]$$
describes the probability of receiving joint observation $\mathbf{o}$ when the joint action $\mathbf{a}_t$ is performed at state s (alternatively: $O_s^\mathbf{o} = \mathcal{P}[\mathbf{O}_{t+1} = \mathbf{o} | S = s]$.)
 - $\mathfrak{B}_0 = \{\beta_0^i\}_{i=1}^n$ is the initial joint belief, specifying for each agent i the probability distribution over states such that $\beta_0^i(s)$ stands for the probability agent i associates with s being the true initial state.

Distributed AI - centralized design - a single body that can influence the preferences of all agents in the system.

Multi-agent AI - individual agents don't have a group sense of utility.



Dimensions of Multi-Agent Systems

- Control
- Decision-making approach
- Communication
- Observability
- Shared resources/interaction
- Objectives/utilities

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Dimensions of Multi-Agent Systems

- **Control:** from centralized to decentralized.
- **Decision-making approach:** planning, learning, RL
- **Communication:** from no communication to limitless communication
- **Observability:** no sensing (conformant) to full information
- **Shared resources/interaction:** space, energy, etc.
- **Objectives/utilities:** from collaborative to adversarial

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Autonomous Vehicles

- Control - Centralized/Decentralized/Hierarchical/Hybrid?
- Communication - Implicit/Passive Action
Recognition/Explicit?
- Objectives
- Shared Resources
- Observability - What information is available to every vehicle?

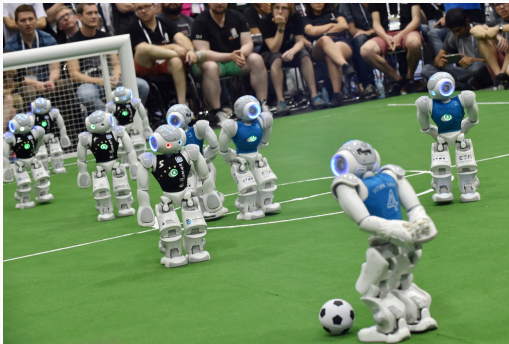


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Robotic soccer

- Control - Centralized/Decentralized/Hierarchical/Hybrid?
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Multi-Agent AI

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Search and Rescue

- Control - Centralized/Decentralized/Hierarchical/Hybrid?
- Communication - Implicit/Passive Action Recognition/Explicit?
- Objectives
- Shared Resources
- Observability - What information is available to every agent?



Human-robot Collaboration

- Control - Centralized/Decentralized/Hierarchical/Hybrid?
- Communication - Implicit/Passive Action Recognition/Explicit?
- Objectives
- Shared Resources
- Observability - What information is available to each agent?



Movie: Amazon Warehouses



- Control - who controls the robots?
- Decision-making approach - how robots choose their actions?
- Communication - how robots communicate with each other?
- Shared resources - what resources are common to all robots?
- Objectives - what is the goal of the robots as a group?
- Robustness - how robust is the system to failures?

Architectures for Control

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- **Centralized**
 - Global controller with global view
 - Good for tightly-coupled tasks, efficiency, adversarial
 - Good for small teams or highly structured environments
 - Requires: high bandwidth/computation/sensing (at least for the leader)
- **Middle Ground: Semi-Centralized**
 - Try to approximate the effect of a centralized system
 - Supervisor and Team (supervisor acts as global controller)
 - Hive-based (homebase or rendezvous to deposit information)
 - Role-based coordination (pre-decide responsibilities)
 - When? Communication is available but slow or limited range.
- **Decentralized**
 - No one has a full world view (peer-to-peer system)
 - Independent acting robots (purely local or no communication)
 - Good for large distributed teams (no centralized bottleneck/failure)
 - Often biologically-inspired (swarm intelligence)

Multiple Agents Introduce Multiple Difficulties

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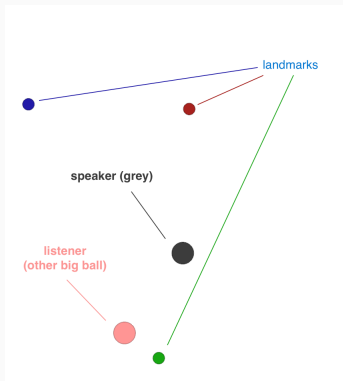
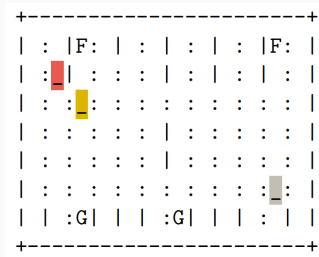
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- Communication
- Collisions
- Coordination
- Task Allocation
- Conflicting Objectives

Example Domains

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Multi-Agent Planning

- Devising effective action policies or strategies for a set of n agents
- In **fully collaborative** settings, the key challenge is coordinating the actions of the individual agents so that the **shared goals** are achieved efficiently
 - Many of the interesting problems in cooperative game theory (such as coalition formation and negotiation) disappear.
 - Rather it becomes more like a standard (one-player) decision problem, where the collection of n players can be viewed as a single player trying to optimize its behavior against nature.

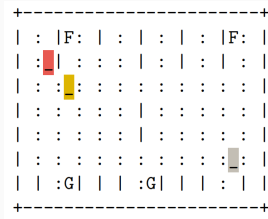
Planning, Learning and Coordination in Multiagent Decision Processes February 1970 Authors: Craig Boutilier Google Inc.

A Market Approach to Multirobot Coordination. Dias et al. 2001

Market-Based Multirobot Coordination: A Survey and Analysis. Dias et al. 2006

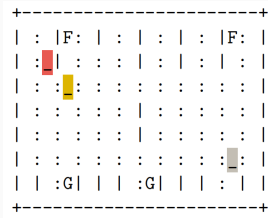
<https://ieeexplore.ieee.org/document/1677943>

Centralized Multi-Taxi

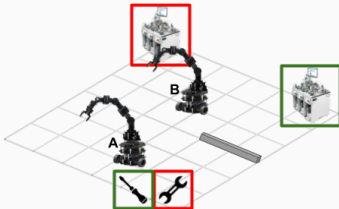


- What is the joint state space and action space?
- What are the assumptions made ? What kind of interactions/ collaborations we support?

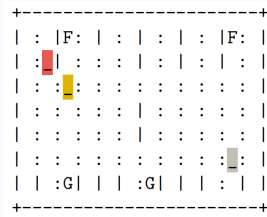
Centralized Multi-Taxi



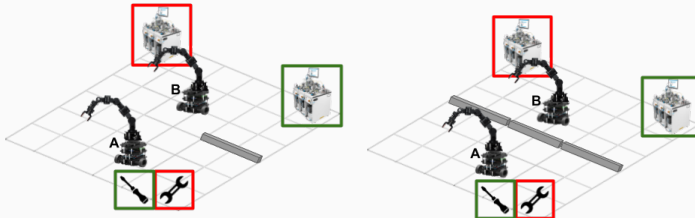
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Centralized Multi-Taxi



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Multi-Agent Planning

- A single agent with joint actions at its disposal allows one to compute (or learn) optimal joint policies by standard methods, provided the agents are of “one mind.”
- However:

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Multi-Agent Planning

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- However:
 - In many settings, agents are self interested
 - may be motivated by non-aligned or conflicting objectives

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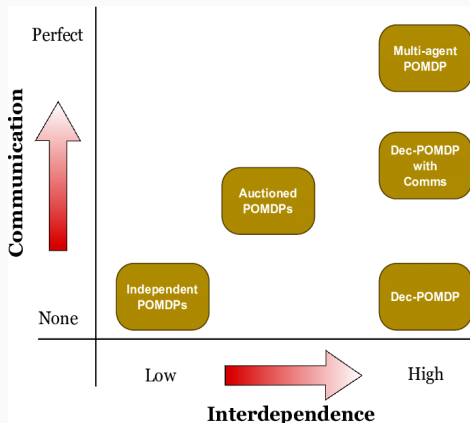
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 - Even in collaborative settings, we expect agents to plan or learn independently, with a limited ability to share information.
 - However, choices made separately may be jointly suboptimal.
 - Need methods for coordination: we must ensure that the individual decisions made can be coordinated so that joint optimality is achieved

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 - Even in collaborative settings, we expect agents to plan or learn independently, with a limited ability to share information.
 - However, choices made separately may be jointly suboptimal.
 - Need methods for coordination: we must ensure that the individual decisions made can be coordinated so that joint optimality is achieved
 - Even in fully-collaborative settings optimal coordination is computationally difficult:
 - best known algorithms are exponential in complexity.
 - approaches striving to compute globally optimal solutions become intractable for teams larger than a few

- Solutions to the coordination problem can be divided into four general classes:
 - **communication-based** (e.g., agents might communicate in order to determine task allocation)
 - **convention-based** (e.g., social laws might be imposed by the system to avoid collisions)
 - **learning-based** (e.g., coordinated policy learned through repeated interaction)
 - **market-based architecture** (e.g., form centralized sub-groups to improve efficiency)

Multi-Agent Planning



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https:

*//www.researchgate.net/publication/221456711_
Decentralized_decision_support_for_an_agent_
population_in_dynamic_and_uncertain_domains*